

Chapter 10

Plant Bugs (Miridae)

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Abstract The Miridae, or plant bugs, with more than 1,300 genera and 11,130 described species, represent the largest and most diverse family of Heteroptera. Although many plant bugs are phytophagous and may cause serious damage to crops, a great many are predatory and may be important in biological control programs. In the Neotropics, about 561 genera and more than 3,400 species are included in seven of the world's eight recognized subfamilies. In this chapter, we give an overview of the classification, general life history, and economic importance and provide identification keys to the Neotropical subfamilies and tribes. For each subfamily, we present a brief diagnosis, an overview of the respective tribes and subtribes, and key literature.

10.1 Introduction

The classification of the Miridae was first outlined by Reuter (1910), who recognized nine subfamilies and 18 divisions. Van Duzee (1917), in his catalog of the Hemiptera of North America north of Mexico, modified Reuter's names and arrangement and listed seven subfamilies. Knight (1923) recognized nine subfamilies based on studies of the arolia (parempodia), pretarsi, and genitalic structures, a grouping similar to that used by Van Duzee (1917). Carvalho (1952) published a synopsis of the higher classification of the family in which he listed six subfamilies, not including Isometopinae, and 24 tribes. Schuh (1976), in evaluating

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pretarsal structures and femoral trichobothria, presented a cladistic analysis that supported a novel six-subfamily arrangement, including the Psallopinae and Isometopinae, but with the Deraeocorinae merged with the Mirinae and the Orthotylinae merged with the Phylinae. Although generally accepted by European hemipterists, Schuh's inclusion of the Isometopidae as a subfamily of Miridae marked the first time that this arrangement had been accepted in the Western Hemisphere. Schuh et al. (2009) recognized seven subfamilies, omitting the Psallopinae, and Jung and Lee (2012), likewise, supported a seven-family classification (minus Psallopinae) based on molecular sequence data, although they considered the Bryocorinae polyphyletic with the bryocorines (Bryocorini) and dicyphines (Dicyphini) having widely separated positions on their cladogram. Carvalho's (1955a) keys to the genera of the world and first modern world catalog (Carvalho 1957, 1958a, b, 1959, 1960) stimulated major interest in the family, and Schuh's (1995) much-needed update was followed by an invaluable online version (Schuh 2013) that has given unprecedented access to information on the Miridae throughout the world. Currently, eight subfamilies (Bryocorinae, Cylapinae, Deraeocorinae, Isometopinae, Mirinae, Orthotylinae, Phylinae, and Psallopinae) are recognized (Henry 2009; Schuh 2013).

The Miridae, commonly called plant bugs (United States and Canada), capsids (Britain), *Blindwanzen* (Germany), and *blomstertaeger* (Denmark), represent the largest family of Heteroptera (Henry and Wheeler 1988; Schuh and Slater 1995; Wheeler 2000a; Henry 2009) and one of the 20 most diverse families of Insecta (Cassis and Schuh 2012), with more than 11,130 described species (Schuh 2013). Henry and Wheeler (1988) suggested that the world fauna will approach 20,000 once the Neotropical species are fully studied. Plant bugs are found in all major zoogeographic regions of the world, except the Antarctic, with centers of diversity in tropical and Mediterranean ecosystems (Wheeler 2001; Cassis and Schuh 2012; Schuh 2013).

The Miridae form a monophyletic group (Wheeler et al. 1993; Wheeler 2001; Schuh et al. 2009; Jung and Lee 2012) characterized by the presence of femoral trichobothria, the loss of nymphal dorsal abdominal scent glands between segments 4 and 5, asymmetrical parameres, two- or three-segmented tarsi, small pulvilli attached near the bases of the claws, a triangular cuneus at the apex of the corium, and a hemelytral membrane with one or two closed cells (Schuh et al. 2009).

The fossil Miridae are insufficiently studied, and although a large number of species have been named (Wheeler 2001), most of them need revision (Zherikhin 2002; Grimaldi and Engel 2005). The earliest fossil mirid is recorded from the Upper Jurassic (Kukalová-Peck 1991; Shcherbakov and Popov 2002; Yao et al. 2007; Jung and Lee 2012). Leston (1979) suggested that the primary speciation of the Miridae took place during the Upper Cretaceous or Lower Cenozoic, following the rise of the angiosperms to dominance (Crepet 1979; Wheeler 2001). Based on molecular data, using the oldest available fossils as calibration points, Jung and Lee (2012) hypothesized that the earliest mirids may have evolved as early as the Permian.

10.2 General Characteristics and Diagnosis (Fig. 10.1)

Many plant bugs are often brightly colored red, orange, and yellow, often with spots and stripes, but the majority are more cryptically colored, often mottled green, gray, brown, or black, blending in remarkably well with their surroundings (Henry and Wheeler 1988; Wheeler and Henry 2004; Henry 2009). Myrmecomorphy is common in the family (Henry 2009), especially in the subfamilies Mirinae (e.g., Herdoniini), Orthotylinae (e.g., Ceratocapsini), and Phylinae (e.g., Cremnorrhini and Pilophorini). The adults vary considerably in size and appearance. They range

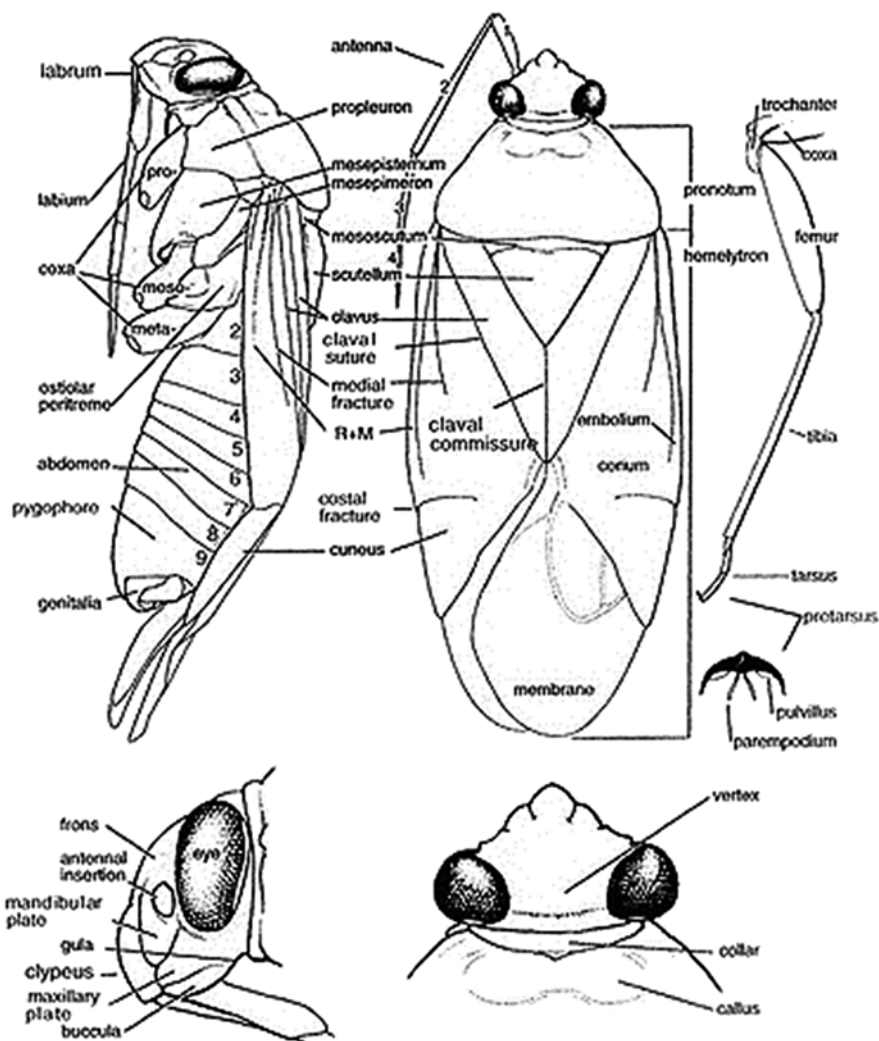


Fig. 10.1 Miridae general morphology (After Schwartz and Footitt 1992)

from little more than one mm, as in certain species of *Tytthus* Fieber, to 15 mm or more in some Neotropical Restheniini. They are elongate to ovoid. The head is triangular and usually prognathous. The eyes are usually large, and ocelli, considered plesiomorphic, are present only in the subfamily Isometopinae. The antennae and labium are four segmented. Most species are fully winged (macroptery), though submacroptery, brachyptery, microptery, and coleoptery are widespread, as are intraspecific wing polymorphism and sexual dimorphism. The hemelytra are distinctive, with the corium divided distally into a triangular cuneus and an apical membrane with one or two unequal cells divided by a short longitudinal vein, a character that is apomorphic for mirids. Trochanters of all legs are divided. The middle and hind femora possess two to 10 trichobothria on the ventral and lateral surfaces. The tarsi are most often three segmented. The pretarsus includes the claws, which can be simple or toothed, with setiform or fleshy parempodia and commonly with fleshy pulvilli and pseudopulvilli. The metathoracic scent glands are paired, and the external efferent or evaporative system is usually well developed. The evaporatorium occupies less than half of the metepisternum with a variably shaped and often diagnostic ostiolar peritreme. The metathoracic spiracle is either exposed and teardrop shaped or recessed. The male genitalia are asymmetrical, including the parameres and endosoma. The endosoma is partially membranous, often inflatable, and is confined by a sclerotized phallosome. The female ovipositor is lacinate; the sperm storage organ is a large, saclike structure. Male and female genitalia have characters important in inferring higher-group relationships and identifying species (Davis 1955; Kelton 1959; Schuh 1975, 1976; Fontes 1981, 1989, 1996; Henry and Wheeler 1988; Schuh and Slater 1995; Wheeler 2001; Cassis and Schuh 2012).

10.3 General Biology and Ecology

Plant bugs are ecologically successful because of their high species diversity, large populations, and wide geographic ranges (Wheeler 2001). Various mirids have distributions that correspond with long-recognized biogeographic regions and intercontinental patterns (e.g., Gondwanan or Paleotropical) (Schuh and Slater 1995). Biogeographic studies have identified centers of endemism, interrelationships, and areas of species richness correlated with areas with high endemism (Wheeler 2001).

Mirids are abundant across a variety of habitats, ranging from mountain tops to salt marshes and coastal dunes (Southwood and Leston 1959; Marples 1966; Denno 1977; Wheeler 2001). Many species specialize in glandular-hairy and even carnivorous plants (Wheeler 2001), and species of *Ranzovius* Reuter are among the few heteropterans that live as commensals in spider webs (Wheeler and McCaffrey 1984; Henry 1999).

The habits and biology of the Miridae have been documented in numerous, widely scattered publications (Henry 2009). Wheeler (2000a, b, 2001) synthesized data on the biology, hosts, and habits, significantly increasing knowledge of the family. Much of what we know about courtship and oviposition behavior, fecundity,

longevity, and voltinism is the result of studies involving species of agricultural importance. The life histories of most plant bugs, however, are unknown.

Miridae may have a significant economic impact on agricultural crops. Many species are major pests (Wheeler 2000a, 2001; Henry 2009; Cassis and Schuh 2012), whereas others are predatory and considered beneficial and have great potential as biological control agents (Wheeler 2000b; Sanchez et al. 2004; Cassis and Schuh 2012). Because even small mirid populations can cause severe injury, applied entomologists frequently underestimate the importance of plant bugs. Feeding symptoms in the Miridae probably are more diverse than in any other heteropteran family (Wheeler 2000a). Most crop plants have at least one mirid species among their complement of pests. Changes in agricultural practices have been responsible for increasing the host range of many plant bugs, and the introduction of numerous exotic plants has increased the supply of new hosts and has allowed the establishment of nonnative taxa around the world (Wheeler 2001).

Mirids often are not recognized as predators. Species with predatory habits do not exhibit the typical specialized morphological characteristics for capturing prey. Predation by mirids on arthropod eggs and early instars of other arthropods make them potentially important in biological control (Wheeler 2000b). Even plant bugs that are significant crop pests can, under certain conditions, be considered beneficial because they are able to switch from phytophagy to zoophagy (Wheeler 2000a, b, 2001). A mirid that is normally a key crop pest can become an important predator of other pests on the same crop. Some species have been described either as pests or as useful natural enemies, depending on where they occur, seasonality, hosts, control measures, and availability of alternative food sources (Wheeler 1976, 2000b, 2001).

Their piercing-sucking mouthparts and the presence of powerful salivary secretions allow mirids to explore nutrient resources, such as dried carrion and feces (Wheeler 2001). Most plant bugs are oligophagous, occurring on members of one plant family or on plants belonging to a few closely related families (Wheeler 2000a). Only a few species are considered monophagous (Wheeler 2001). For example, Pires et al. (2007, 2008) showed that *Platyscytus decempunctata* (Carvalho) feeds only on *Solanum cernuum* Vell. (Solanaceae). In contrast, some species are strongly polyphagous, such as *Lygus lineolaris* (Palisot de Beauvois), which may feed on more than 300 plants (Young 1986).

10.4 Life History

Mirids often have life cycles that last no more than six weeks. They usually have five nymphal instars, although the number may be as low as three or as high as six (Schuh and Slater 1995; Wheeler 2001). They reach the adult stage in 15–30 days. Eggs are usually inserted into plant tissue. In most temperate regions, mirids are univoltine and overwinter in the egg stage. Others, however, such as *Lygus* spp., have multiple generations per year and often overwinter as adults. Eclosion occurs in early to late spring in phytophagous species, which are associated with new

growth and flower development of the host plant. Predatory species often have life cycles synchronized with those of their preferred prey and usually appear as adults later in the season than phytophagous species. The life histories of most tropical Miridae, however, are poorly studied or are unknown (Schuh and Slater 1995; Wheeler 2001).

Plant bugs are remarkable for their genitalic evolution (Kelton 1959; Cassis and Schuh 2012), including traumatic insemination and sexual antagonistic coevolution (Tatarnic et al. 2006; Tatarnic and Cassis 2010; Cassis and Schuh 2012). Parthenogenesis occurs in a few mirid species, such as the predatory *Campyloneura virgula* (Herrich-Schaeffer), where males are rare (Wheeler and Henry 1992; Henry 2012a).

The Miridae have significant potential as indicators of ecological changes, mainly because they are susceptible to many pesticides and are vulnerable to habitat disturbances (Fauvel 1999; Wheeler 2001). Brown (1991) presented a list of mirids as potential bioindicators in conservation biology studies. Taking into account their great species diversity, wide range of feeding habitats, and the damage they inflict on fruits, field crops, and ornamental plants, the Miridae are among the most important of all insect groups (Henry and Wheeler 1988; Wheeler 2001; Cassis and Schuh 2012).

10.5 Classification and Diversity

The Miridae, placed (with Thaumastocoridae and Tingidae) in the superfamily Miroidea, are currently separated into eight subfamilies: Bryocorinae, Cylapinae, Deraeocorinae, Isometopinae, Mirinae, Orthotylinae, Phylinae, and Psallopinae (Schuh 2013). More than 11,130 described species are included in 44 tribes worldwide (Schuh 2013; Schuh and Menard 2013).

All mirid subfamilies represented in the Neotropics (considering all of Mexico, Central and South America, and the West Indies), except the Psallopinae (Wolski and Henry 2015). About 3,429 species and 561 genera are included in 25 tribes, representing 27 % of the world mirid fauna. Mexico has about 621 species; Central America, 440; South America, 2,008; and the West Indies, 406. Tables 10.1 and 10.2 show countries of the Neotropical Region with their subfamilies, tribes, number of genera, and species represented. The countries with the largest number of genera and species are Brazil (280 genera, 1,081 species), Mexico (213, 621), Argentina (163, 524), Peru (124, 327), and Panama (112, 234). For the largest islands of the West Indies (Hernández and Henry 2010), Cuba has 57 genera and 105 species recorded; Puerto Rico, 40 and 63; Jamaica, 36 and 48; Hispaniola, 22 and 28; and Grenada, 15 and 23. Mexico has 20 tribes; Central America, 23; and South America, 22. The majority of the genera are represented by only a few species (Fig. 10.2); 49 % percent have only one species; 62 %, one or two species; and 75 %, one to four species.

The Neotropical genera with the greatest numbers of species in descending order are *Prepops* Reuter (Mirinae), *Phytocoris* Fallen (Mirinae), *Eccritotarsus* Stål (Bryocorinae), *Ceratopsus* Reuter (Orthotylinae), *Taedia* Distant (Mirinae), *Orthotylus* Fieber (Orthotylinae), *Pycnoderes* Guérin-Menèville (Bryocorinae), *Lopidea* Uhler (Orthotylinae), *Fulvius* Stål (Cylapinae), and *Dagbertus* Distant (Mirinae).

The percentages of Neotropical species by subfamily (Fig. 10.3) in descending order are Orthotylinae (33 %), Mirinae (28 %), Deraeocorinae (14 %), Bryocorinae (10 %), Cylapinae (7.4 %), Phylinae (7 %), and Isometopinae (0.50 %). Ninety percent of the Miridae in the Neotropical Region are represented in the following tribes: Orthotylini (34 %) (Orthotylinae), Mirini (19 %) (Mirinae), Eccritotarsini (10 %) (Bryocorinae), Phylini (8 %) (Phylinae), Deraeocorini (8 %) (Deraeocorinae), Herdoniini (5 %) (Mirinae), Restheniini (3 %) (Mirinae), and Clivinematini (3 %) (Deraeocorinae).

Most of the literature supporting the study of Neotropical Miridae is found primarily in taxonomic papers, including keys; catalogs; reviews of subfamilies, tribes, and genera; and descriptions of genera and species (e.g., Carvalho 1955a, 1957–1960; Schuh 1995, 2013; Carvalho and Costa 1997; Ferreira 1980; Ferreira and Henry 2011).

Maldonado's (1969) Miridae of Puerto Rico and Hernández and Henry's (2010) Miridae of Cuba contain diagnoses and keys to the subfamilies, tribes, genera, and species that are useful in recognizing many of the more widely distributed taxa. Studies on insect-plant relationships, biogeography, and integrated pest management are few, but see literature cited in Ferreira et al. (2001), Ferreira and Henry (2011), Schuh (1991, 2014), and Wheeler (2000a, b, 2001). Carvalho and Carpintero (1993) and Carpintero et al. (2006) provided a checklist of the Miridae of Argentina. Although treating mostly North American taxa, H. H. Knight's (1923, 1941, 1968) comprehensive works on the Miridae of Connecticut, Illinois, and the Nevada Test Site are still useful for keying subfamilies, tribes, and genera.

There also are few Neotropical mirid studies hypothesizing phylogenetic relationships. Ferreira (1998) revised the tribe Clivinematini and gave a phylogenetic analysis of the tribe; Schuh (1991) treated the Pilophorini; Schwartz (1987, 2008), the Stenodemini; and Tatarnic and Cassis (2012), the Halticini. Forero (2008) revised and provided an analysis of the *Hadronema* Uhler group; Henry and Kim (1984), *Neurocolpus* Reuter; Henry (1991, 1999, 2012b), *Keltonia* Knight, *Pseudatomoscelis* Reuter, *Ranzovius* Distant, and *Tyththus* Fieber; Sadowska-Woda et al. (2008), *Fulvius* Stål (based on molecular data); Schuh and Schwartz (1985), *Rhinacloa* Reuter; and Schwartz (1984, 2011), *Irbisia* Reuter and *Slaterocoris* Wagner.

One cannot consider the Miridae without mentioning José C. M. Carvalho. He published a majority of the taxonomic work on the Neotropical Miridae in an amazing, career-long series of nearly 500 papers in which he described almost 400 genera and more than 2,100 species (Schuh 2013). Carvalho and Froeschner (1987, 1990, 1994) listed the genera and species described by Carvalho from 1943 to January 1993, along with their type depositories.

Table 10.2 Number of genera and species by tribe in countries of South America

South America		COL		VEN		GUY		SUR		FGU		ECU		PER	
		GN	SP	GN	SP	GN	SP	GN	SP	GN	SP	GN	SP	GN	SP
Subfamily	Tribe														
Bryocorinae	Bryocorini	0	0	0	0	0	0	0	0	0	0	1	1	1	1
Bryocorinae	Dicyphini	3	9	2	3	1	2	3	5	1	2	4	12	6	13
Bryocorinae	Eccritotarsini	7	22	15	24	2	2	6	8	2	3	9	19	18	51
Cylapinae	Cylapini	0	0	2	2	0	0	0	0	0	0	2	2	1	6
Cylapinae	Fulvini	2	6	2	9	2	3	2	4	2	6	2	7	2	5
Cylapinae	Vanini	1	1	0	0	0	0	0	0	0	0	0	0	0	0
Deraeocorinae	Clivinematini	2	3	3	4	0	0	0	0	0	0	1	1	3	4
Deraeocorinae	Deraeocorini	4	6	2	3	0	0	0	0	1	1	1	1	3	7
Deraeocorinae	Hyaliodini	4	12	5	6	1	1	1	1	0	0	6	7	10	16
Deraeocorinae	Surinamellini	0	0	1	1	0	0	1	1	0	0	1	1	1	1
Deraeocorinae	Termtatophylini	0	0	1	2	1	1	1	1	0	0	0	0	0	0
Isometopinae	Diphlebiini	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Isometopinae	Myiommini	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Mirinae	Herdoniini	1	1	1	1	0	0	1	1	0	0	2	2	7	7
Mirinae	Mirini	20	46	19	39	3	3	9	10	12	28	25	68	38	96
Mirinae	Restheniini	7	17	4	9	0	0	1	3	1	2	5	14	7	51
Mirinae	Stenodemini	3	8	3	3	0	0	1	1	0	0	4	6	5	9
Orthotylinae	Ceratocapsini	0	0	1	3	1	1	1	2	0	0	1	1	1	3
Orthotylinae	Halticini	0	0	0	0	0	0	1	2	0	0	1	2	1	2
Orthotylinae	Orthotylini	7	13	7	14	1	1	1	1	3	3	8	11	10	20
Phyllinae	Hallodapini	1	1	0	0	0	0	0	0	0	0	0	0	0	0
Phyllinae	Phyllini	7	8	8	15	0	0	5	6	1	1	10	22	9	32
Phyllinae	Pilophorini	0	0	0	0	1	1	0	0	1	1	0	0	1	3
	Total	69	153	76	138	13	15	34	46	24	47	83	177	124	327

Subfamily	Tribe	BOL		PAR		ARG		URU		CHI		BRA	
		GN	SP	GN	SP	GN	SP	GN	SP	GN	SP	GN	SP
Bryocorinae	Bryocorini	0	0	1	1	1	1	0	0	0	0	1	3
Bryocorinae	Dicyphini	1	4	4	4	4	10	1	2	2	3	5	27
Bryocorinae	Ecritotarsini	12	29	8	13	18	53	3	5	2	2	34	152
Cylapinae	Cylapini	2	2	0	0	1	1	0	0	0	0	12	23
Cylapinae	Fulvini	2	3	2	9	3	15	0	0	1	2	9	32
Cylapinae	Vanini	1	1	0	0	0	0	0	0	0	0	1	1
Deraeocorinae	Clivematini	0	0	0	0	6	9	0	0	0	0	4	10
Deraeocorinae	Deraeocorini	0	0	1	1	4	7	0	0	4	6	11	23
Deraeocorinae	Hyaliodini	6	6	3	4	7	13	0	0	2	2	11	38
Deraeocorinae	Surinamellini	0	0	0	0	1	2	0	0	0	0	4	9
Deraeocorinae	Teratophyllini	0	0	0	0	0	0	0	0	0	0	1	1
Isometopinae	Diphebinini	0	0	0	0	0	0	0	0	0	0	0	0
Isometopinae	Myiommini	0	0	0	0	3	7	0	0	0	0	1	3
Mirinae	Herdoniini	2	2	7	11	10	27	1	1	0	0	20	54
Mirinae	Mirini	20	34	20	39	31	123	5	5	10	44	55	277
Mirinae	Restheniini	6	19	7	22	15	96	2	5	3	6	17	134
Mirinae	Stenodemini	2	2	3	3	10	26	3	4	1	2	10	20
Orthotylineae	Ceratocapsini	1	4	1	4	1	17	0	0	0	0	1	47
Orthotylineae ^a	Halticini	0	0	0	0	1	3	1	1	0	0	1	3
Orthotylineae	Orthotylini	8	13	8	13	30	79	1	1	10	33	55	145
Phyllinae ^b	Hallodapini	0	0	0	0	1	1	0	0	0	0	1	1
Phyllinae	Phyllini	4	6	3	4	15	31	0	0	8	16	24	72
Phyllinae	Pilophorini	0	0	1	1	1	3	0	0	0	0	1	7
	Total	67	125	69	129	163	524	17	24	43	116	279	1,082

COL Colombia, *VEM* Venezuela, *GUY* Guyana, *SUR* Suriname, *FGU* French Guiana, *ECU* Ecuador, *PER* Peru, *BOL* Bolivia, *PAR* Paraguay, *ARG* Argentina, *URU* Uruguay, *CHI* Chile, *BRA* Brazil

^aExcluding Coridromini

^bPhyllini (*sensu lato*), prior to Schuh and Menard (2013) and Menard et al. (2014)

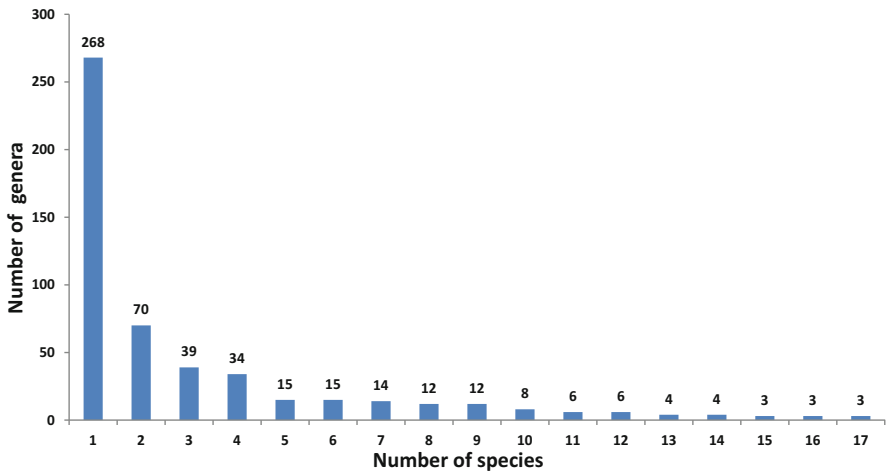


Fig. 10.2 Number of genera in relation to number of species of Neotropical Miridae

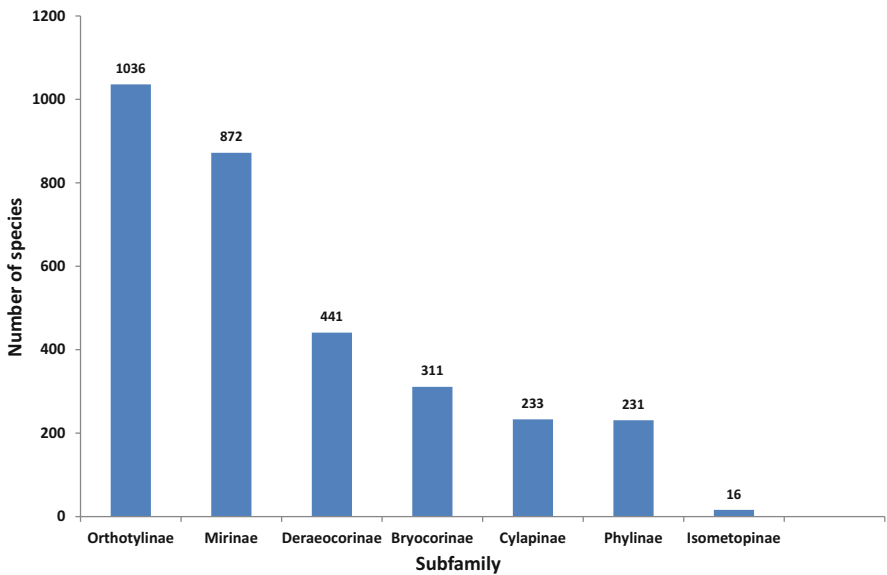


Fig. 10.3 Number of species by subfamily of Neotropical Miridae

In the following section, we provide keys to the subfamilies and tribes of Neotropical Miridae, modified from Schuh and Slater (1995), Carvalho and Costa (1997), and Hernández and Henry (2010), and other references as noted, followed by a diagnosis of the subfamily and a highlight of the most significant papers containing substantial numbers of new taxa and/or identification keys. Only seven of the eight recognized subfamilies are included. Previously, the Neotropical genus

Isometocoris Carvalho and Sailer had been considered the only New World representative of the subfamily Psallopinae (Henry and Maldonado 1982). In a forthcoming paper, however, Wolski and Henry (2015) document that *Isometocoris* is not a psallopine; thus, this subfamily is omitted from the key.

Key to the Neotropical subfamilies of Miridae

1. Tarsi two segmented, the first segment shorter than the second; ocelli..... present Isometopinae
- Tarsi two or three segmented; ocelli absent 2
2. Parempodia large, fleshy, apically convergent or divergent, arising between claws; phallosome never cone shaped..... 3
- Parempodia setiform, if large, fleshy, and convergent, then with a cone-shaped phallosome nested within the left paramere and protruding posteriorly from genital capsule..... 4
3. Parempodia divergent at apices, usually thickened; a prominent collar present, separated from anterior margin of pronotum by a deep furrow present..... Mirinae
- Parempodia convergent at apices, usually more slender; pronotal collar absent, if present, not separated from pronotum by a deep suture Orthotylinae
4. Pulvilli present, free, or connected along inner surface of claws, sometimes difficult to see, in which case a collar is always absent 5
- Pulvilli absent; collar present or absent 6
5. Pulvilli arising from base or inner margin of claw; hemelytral membrane with two cells; tarsi uniformly slender..... Phylinae
- Pulvilli arising from ventral surface of claw; hemelytral membrane with one or two cells; tarsi thickened distally or slender with tiny pretarsi..... Bryocorinae
6. Claws distinctly toothed or thickened at bases; never toothed apically Deraeocorinae
- Claws not toothed or thickened basally; apex almost always with a fine subapical tooth Cylapinae

10.5.1 Subfamily Bryocorinae

This subfamily is recognized by the three-segmented tarsi, the often large and fleshy pulvilli (Eccritotarsini) attached to the bases of the claws and the often distally thickened (Eccritotarsini) or sometimes slender (Dicyphini) tarsal segments, with tiny indistinct claws. The Bryocorini and Eccritotarsini have only one closed cell on the hemelytral membrane, whereas the Dicyphini (Dicyphina and Monaloniina) have two.

Carvalho and Ferreira (1995) keyed 66 of the Neotropical genera of Bryocorinae (Eccritotarsini), and Ferreira and Henry (2011) provided a synopsis and keys to the four tribes, 24 genera, and 56 species of Minas Gerais, Brazil.

The following key is modified from Schuh and Slater (1995), Hernández and Henry (2010), and Ferreira and Henry (2011).

Key to the Neotropical tribes of Bryocorinae

1. Hemelytral membrane with two closed cells; tarsi uniformly slender; delicate slender species, sometimes spotted dorsally; genital capsule often bearing tubercles or processes on the right side; pretarsal structures very small, pulvilli small; parempodia present (includes Dicyphina and Monaloniina) Dicyphini
- Hemelytral membrane with only one closed cell; tarsi dilated distally; oval, robust species, without dorsal spots; genital capsule without tubercles on the right side; pulvilli large, distinct; parempodia present or absent..... 2
2. Claws without true pulvilli, but with pseudopulvilli arising from the unguitractor plate, recurved and convergent apically; parempodia absent; metathoracic scent efferent system with bulbous peritremal disc and distinct ostiolar canal Bryocorini
- Claws with large and flattened pulvilli, arising or continuing from the inner claw surface, without pseudopulvilli; parempodia asymmetrical; metathoracic scent efferent system usually with a narrow tongue-shaped peritremal disc and weakly developed ostiolar canal Ecritotarsini

Bryocorini (Fig. 10.4)

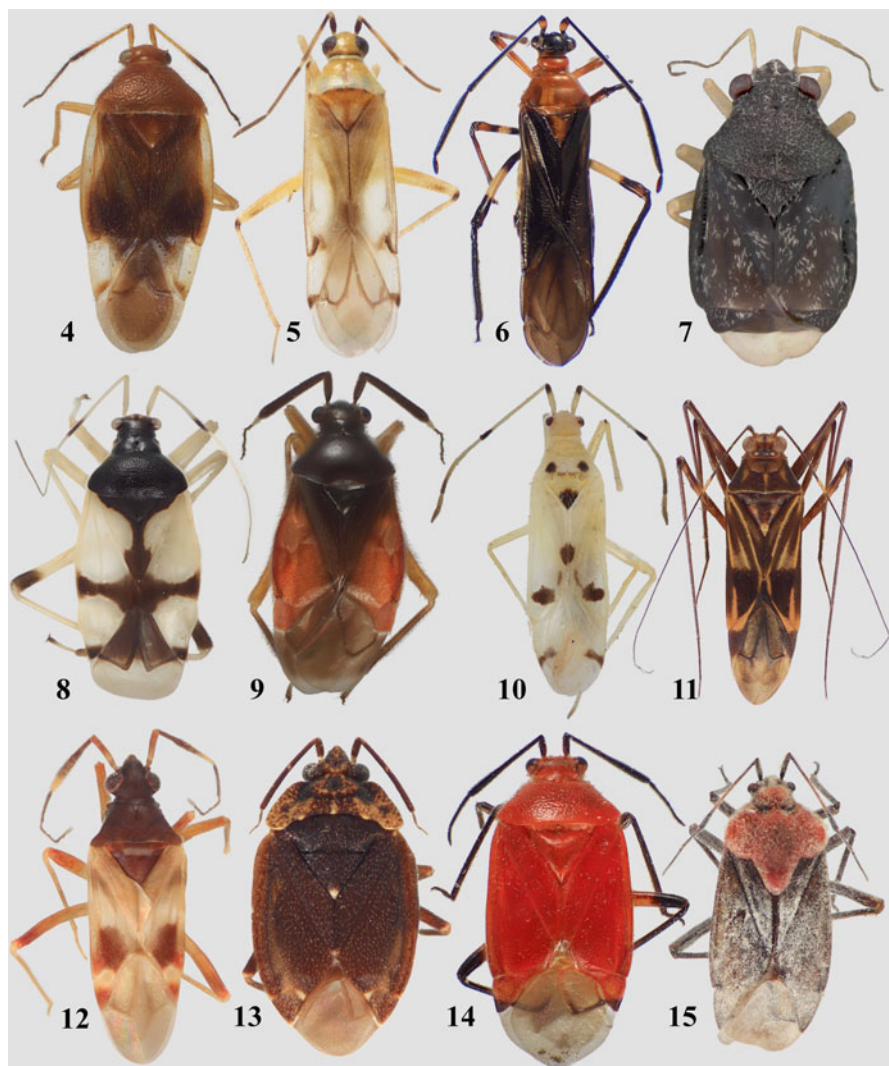
The Bryocorini are the smallest of the three tribes, containing four genera in the world, with only *Monalocoris* Dahlbom occurring in the Neotropics. These small, oval bugs are characterized by the rounded pronotal collar, reduced number of femoral trichobothria, and their specialized feeding on ferns (Pterophyta). Only seven Neotropical species are known (Schuh 2013).

Dicyphini (Figs. 10.5 and 10.6)

The Neotropical Dicyphini are recognized by their slender body form, elongate head, slender antennae, trapeziform pronotum with a distinct narrow collar, wide mesoscutum, translucent hemelytra, often reduced auricles and evaporative area, and respiratory horns on the egg.

The subfamily is separated into three subtribes, of which only the Dicyphina Monaloniina occur in the Neotropics. Cassis (1984) revised and keyed Dicyphina and gave generic status to many of the previous subgeneric names, such as *Engytatus* Reuter, *Nesidiocoris* Kirkaldy, and *Tupiocoris* China and Carvalho. Carvalho and Ferreira (1994b) keyed the Neotropical genera.

The widespread *Engytatus modestus* (Distant) and *Nesidiocoris tenuis* (Reuter) are well-known pests of various crop plants, including tobacco, tomato, and eggplant



Figs. 10.4–10.15 Neotropical Miridae. 4, *Monalocoris pallidiceps* (Reuter) (Bryocorinae: Bryocorini). 5, *Nesidiocoris tenuis* (Reuter) (Bryocorinae: Dichypini: Dicyphina). 6, *Monalonion annulipes* Signoret (Bryocorinae: Dicyphini: Monaloniina). 7, *Cyrtocapsus femoralis* Reuter (Bryocorinae: Eccritotarsini). 8, *Eccritotarsus brevicuneatus* Carvalho (Bryocorinae: Eccritotarsini). 9, *Pachymerocerus fairmairei* (Stål) (Bryocorinae: Eccritotarsini). 10, *Chius maculatus* Distant (Bryocorinae: Eccritotarsini). 11, *Cylapus striatus* Reuter (Cylapinae: Cylapini). 12, *Fulvius anthocoroides* Stål (Cylapinae: Fulviini). 13, *Peritropis izyai* Wolski and Henry (Cylapini: Fulviini). 14, *Ambracius dufouri* Stål (Deraeocorinae: Clivinematini). 15, *Clivinema regalisimilis* Carvalho (Deraeocorinae: Clivinematini) (Photos by G Ouellette)

(Tanada and Holdaway 1954; El-Dessouki et al. 1976; Wheeler 2000a, 2001), but both species also prey on aphids, mealybugs, lepidopteran eggs, and other small arthropods (Wheeler 2000b), becoming useful predators, especially in greenhouses.

The subtribe Monaloniina, containing 21 genera worldwide, is represented in the New World only by the genus *Monalonion* Herrich-Schaeffer. Carvalho (1972) reviewed the genus and provided a key to 11 of the 16 known species. Monaloniines are relatively large, shiny bugs, with laterally produced eyes and distinct pseudopulvilli. Several species of *Monalonion* have been reported damaging the developing fruits and pods of *Theobroma cacao* L. (Malvaceae) (de Abreu 1977; Wille 1944). Torres Jaimes et al. (2012) observed *M. velezangeli* Carvalho and Costa causing serious injury to the buds, flowers, and fruits of avocado, *Persea americana* Mill. (Lauraceae) in Colombia.

Eccritotarsini (Figs. 10.7, 10.8, 10.9, and 10.10)

Of the approximately 107 genera and 628 species of Eccritotarsini known in the world, 70 genera and about 452 species occur in the Neotropics (Schuh 2013). The tribe is recognized by the swollen femoral trichobothrial bases, the fleshy pulvilli attached to the inner surface of the claw, the asymmetrical parempodia, the dilated tarsi, the greatly reduced metathoracic scent gland opening, the single closed membranous cell, and the strongly developed male genitalia (Stonedahl 1988; Ferreira and Henry 2011).

Carvalho and Ferreira (1995) keyed 66 of the Neotropical genera, and Ferreira and Henry (2011) keyed the 18 genera and 42 species known from Minas Gerais, Brazil.

Other important contributions include Carvalho's (1953) revision and key to the species of *Eccritotarsus* Stål; Schaffner and Carvalho's (1981) review of *Hesperolabops* Kirkaldy; Carvalho and Schaffner's (1985) descriptions of numerous new species of *Adneella* Carvalho, *Neella* Reuter, *Sysinas* Distant, *Neofurius* Distant, *Neosilia* Distant, *Nototremates* Carvalho and China, and *Dichroocoris* Reuter; and Henry and Carvalho's (1987) key to the species of *Cyrtocapsus* Reuter. Uceli et al. (2009) reviewed *Pachymeroceroides* Carvalho and Gomes and described the peculiar male hemelytra.

This subfamily contains a number of economically important taxa. *Pycnoderes quadrimaculatus* Guérin-Meneville, often called the bean capsid, is a well-known pest of beans, squash, sweet potatoes, and garden crops, where it causes chlorosis, mottling, leaf drop, and premature aborting of fruits (Wheeler 2000a). Feeding by the Neotropical *Tenthecoris orchidearum* (Reuter) may produce chlorotic spots or even kill the foliage on orchids, and, in heavy infestations, the entire plants may die (Gimingham 1928; Wheeler 2001). In the early twentieth century, species of *Tenthecoris* were considered serious pests of orchids grown in greenhouses in the Britain (Carvalho and Leston 1952), France (Denis 1908), and the United States

(Weiss 1917), but with the advent of modern control practices, these ecritotarsines seldom become a problem (Wheeler 2001).

10.5.2 Subfamily Cylapinae

Of the cylapine tribes recognized by Cassis and Schuh (2012), only Cylapini (12 genera and 42 species), Fulviini (15 genera and 88 species), and Vaniini (one genus and four species) occur in the Neotropics (Gorczyca 2000, 2006; Schuh 2013). This subfamily is recognized by the long labium, often extending well beyond the hind coxae; the long, slender antennae; the membrane having two closed cells; and the long, slender legs, setiform parempodia, lack of pulvilli, and long, slender claws, usually with a distinct subapical tooth. Members of the Cylapini and Vaniini have relatively broad heads, often with stylate or protruding eyes (e.g., *Cylapus* Say), whereas the Fulviini (e.g., *Fulvius* Stål and *Peritropis* Uhler) have more elongate heads, with smaller, nonprotruding eyes.

Gorczyca (2006) cataloged the subfamily for the world, and Gorczyca (2000) and Cassis and Monteith (2006) studied the phylogeny of the group. Gorczyca (2006) recognized four tribes, whereas Cassis and Monteith (2006) and Cassis and Schuh (2012) listed five, with the addition of Vaniini. Carvalho and Ferreira (1994a) provided a key to the 20 Neotropical genera.

The following key is modified from Gorczyca (2000), with information from Cassis and Schuh (2012) incorporated.

Key to the tribes of Cylapinae

1. Body elongate oval; head horizontal, elongate, tapering anteriorly; eyes relatively small, not bulging or stylate; antenna usually shorter than body, second segment longest Fulviini
- Body elongate; head vertical, somewhat flattened in front; eyes bulging, often stylate; antenna usually longer than the body, third segment longest 2
2. Labium short, relatively stout, extending to or only slightly beyond hind coxae; parempodia flattened Vaniini
- Labium long, slender, extending well beyond hind coxae onto abdomen; parempodia rounded Cylapini

Cylapini (Fig. 10.11)

Members of the tribe are best recognized by their stalked eyes, long slender labium, often extending to the genital segments, auriculate evaporatorium, and toothed claws with setiform parempodia.

Twelve genera and 42 species of Cylapini are known in the Neotropics (Schuh 2013). Carvalho and Fontes (1968a) provided a key to six genera of the *Cylapus* complex (*Amapacylapus* Carvalho and Fontes, *Cylapocerus* Carvalho and Fontes, *Cylapus* Say, *Peltidocylapus* Poppius, *Trichocylapus* Poppius, and *Valdasus* Stål), Carvalho (1987a) added two new genera (*Cylapinus* Carvalho, *Tucuruisca* Carvalho) six new species, and Carvalho (1989a) described five genera (*Adcylapocoris* Carvalho, *Cylapocorella* Carvalho, *Cylapocorisca* Carvalho, *Microcylapus* Carvalho, *Valdasoides* Carvalho) and two species.

Fulviini (Figs. 10.12 and 10.13)

Fifteen genera and 88 species of Fulviini are known from the Neotropical Region (Schuh 2013). Carvalho and Costa (1994) revised and keyed the species *Fulvius*, one of the larger Neotropical mirid genera; Ferreira and Henry (2002) added two new species from Brazil, and Hernández and Henry (2010) described one from Cuba; Sadowska-Woda et al. (2008) presented a preliminary phylogenetic analysis, and Wolski and Henry (2012, 2013) revised *Peritropis* Uhler, describing 17 new species. Carvalho (1989a) described a number of new species (*Adcylapocoris* Carvalho, *Cylapocorella* Carvalho, *Cylapocorisca* Carvalho), and Henry and Silva de Paula (2004) described *Rhyparochromomiris femoratus*, a peculiar ant-mimetic species resembling a herdoniine mirine with greatly swollen front femora similar to those found in certain Rhyparochromidae (Lygaeoidea). Although Schuh (2013) included *R. femoratus* in the Cylapini, the long second antennal segment and male parameres suggest that this peculiar cylapine belongs in the Fulviini.

Vaniini

Vaniines are often brightly colored with orange and red markings. Their protruding eyes and flattened parempodia will distinguish them from most other Cylapinae. Gorczyca (1997) treated the *Vannius* complex (*Cylapus* Say, *Valdasus* Stål, *Vannius* Distant) and cataloged the world fauna. Cassis and Monteith (2006) presented a phylogenetic analysis of the *Vannius* complex, and Cassis and Schuh (2012) provided a summary of relationships.

Little is known of the habits of most cylapines. Species of *Fulvius* are often taken under the loose bark of dead trees, and most are thought to be predatory (Wheeler 2001; Henry et al. 2011), though a few observations suggest that some may feed on fungi (Schuh 1976). *Fulvius anthocoroides* (Reuter) has been observed feeding on mites and insects in stored nuts, and the North American *F. imbecilis* Say is documented feeding on Diptera and Coleoptera larvae and other arthropods (Henry et al. 2011). Herring (1976) considered his new genus and species, *Trynecoris lawrencei*,

a predator of ciid beetle larvae associated with fungi growing on forest trees. Though long thought to be predaceous, Wheeler and Wheeler (1994) observed fungal spores and conidia in the gut contents of the North American *Cylapus tenuicornis* (Say) (and a Neotropical *Cylapus* sp.), providing evidence for mycophagous feeding habits in the genus. Cassis and Monteith (2006) noted that their new vaniine genus and species, *Kauakamiris krypton*, described from New Caledonia, was most easily collected in moldy cut-branch traps comprised of no particular plant species, suggesting it is mycetophagous.

10.5.3 Subfamily Deraeocorinae

Members of this subfamily can be distinguished by the distinctly punctate, often polished dorsum and the basally toothed claws, although certain other deraeocorines (e.g., Teratophylini) may lack distinct dorsal punctures and have a dull coloration (Schuh and Slater 1995; Cassis 1995; Hernández and Henry 2010).

The following key is adapted from Carvalho (1955a) and Hernández and Henry (2010).

Key to Tribes of Deraeocorinae

1. Head elongate, pointed apically, with anthocorid-like facies; eyes large, placed far from basal angle of the head..... Teratophylini
- Head not elongate, without anthocorid-like facies; eyes variable, placed near basal angle of the head..... 2
2. Pronotum with a distinct impressed line running from anterolateral corner to margin of calli, sometimes strongly punctate Clivinematini
- Pronotum without an impressed line..... 3
3. Hemelytra hyaline; emboliar margin of corium wide; males and females with an elongate anal tube Hyaliodini
- Hemelytra not hyaline; emboliar margin of corium narrow; males and females without an anal tube 4
4. Species ant mimetic; antennal segments linear; vertex depressed, sometimes punctate; male vesica with two membranous lobes apically Surinamellini
- Species not ant mimetic; antennal segment I thicker than remaining segments, segment II clavate at apex; vertex smooth, not depressed; male vesica without two membranous lobes apically Deraeocorini

Carvalho's (1955a) now outdated keys to the world genera include the most useful key to the Neotropical deraeocorine genera. Maldonado (1969) keyed the genera and species of Puerto Rico, and Hernández and Henry (2010) treated the Cuban genera.

Clivinematini (Figs. 10.14 and 10.15)

Seventeen genera and about 55 species of Neotropical clivinematines are known (Schuh 2013). Henry (1979a) reviewed *Bothynotus* Fieber, and Ferreira and Henry (2010) revised *Ambracius* Stål. Ferreira (1993, 1996a, b, 1998, 2001) provided a comprehensive treatment of the Clivinematini, describing four new genera and providing information on their biogeography and biology, a cladistic analysis, and a key to the 17 Neotropical genera. Ferreira and Schaffner (2004) reviewed *Admetus* Distant, and Kerzhner and Schuh (1998) provided the replacement name *Admetomiris* for the preoccupied *Admetus*. Carpintero et al. (2008) added the new genus *Valdesiana* Carpintero and Dellapé and provided a key to the six genera and nine species of Clivinematini from Argentina.

Based on a few observations, most clivinematines are thought to prey on species of Sternorrhyncha, especially those in the family Ortheziidae. Knight (1928) reported *Clivinema sericea* Knight feeding on a species of *Orthezia* (Hemiptera: Ortheziidae) in New Mexico, Miller and Schuh (1995) observed *C. coalina* Bliven preying on *O. annae* Cockerell on *Atriplex polycarpa* (Torr.) Wats. in California, Ferreira (1998) reported *Hemicerocoris bicolor* Carvalho feeding on soft scales on guava in Mexico, and Ferreira and Henry (2010) documented records of *Ambracius dufouri* Stål preying on *Orthezia* sp. in Brazil.

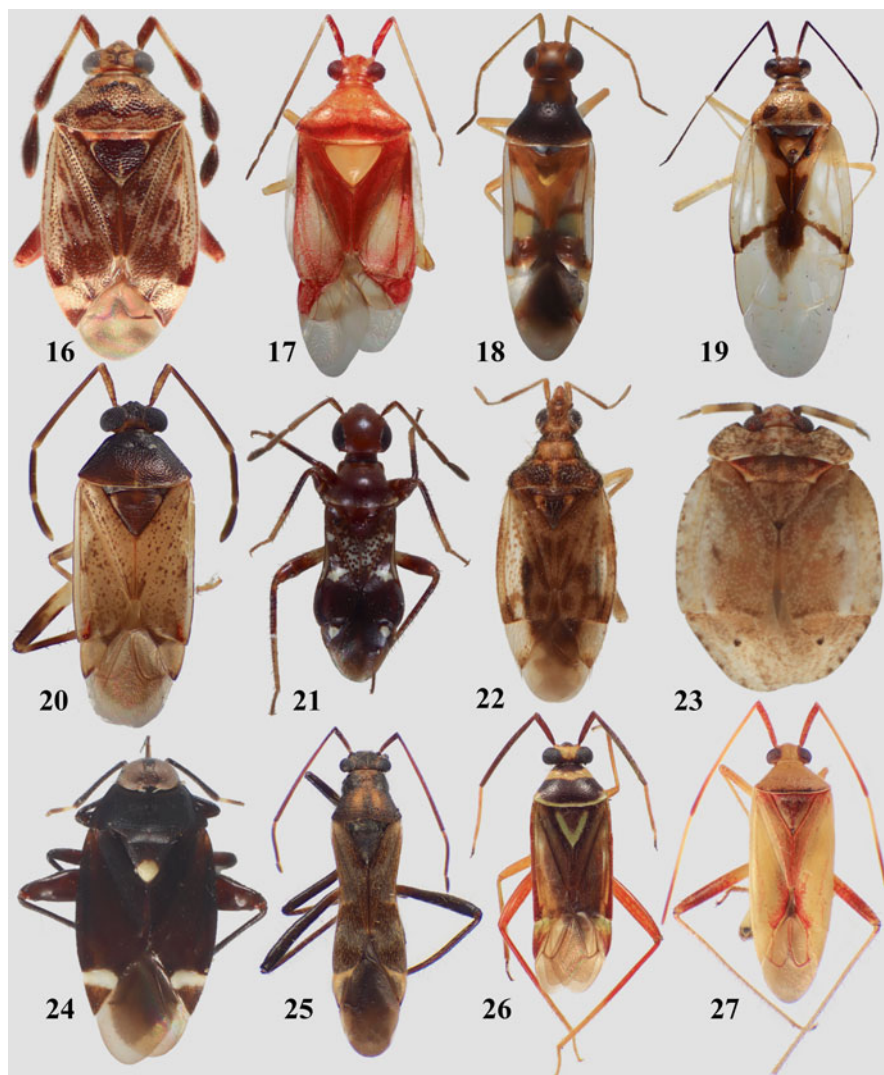
Deraeocorini (Fig. 10.16)

Twenty-seven genera and 62 species of the Deraeocorini are recorded from the Neotropical Region (Schuh 2013). Stonedahl et al. (1997) keyed the North American genera, which include a few of the Neotropical taxa. Most work on the nominotypical tribe has been descriptive. Papers containing keys to species include Carvalho and Maldonado's (1982) study of *Lundiella* and Carvalho and Costa's (1990c) review of *Diplozona* Van Duzee. Hernández and Stonedahl (1999) most recently added the three new genera *Acutifromiris*, *Cephalomiroides*, and *Scutellograndis* from Chile.

Nearly all members of this tribe are considered zoophagous. In North America, Wheeler et al. (1975) showed that *Deraeocoris nebulosus* (Uhler) was an effective predator of various ornamental pests, including mites, aphids, scales, whiteflies, psyllids, and lace bugs. Wheeler (2001) summarized the predatory habits of numerous species of *Deraeocoris* occurring mostly in the Nearctic and Palearctic regions.

Hyaliodini (Figs. 10.17, 10.18, and 10.19)

Twenty genera and about 111 species of Hyaliodini are known from the Neotropics (Schuh 2013). The most important works, including keys to species, are Carvalho and Schaffner (1977) for *Annona* Distant, Carvalho (1982) for *Antias* Distant, Carvalho (1974d) for *Auchus* Distant, and Henry and Ferreira (2003) for *Paracarniella* Henry and Ferreira. Several of the larger hyaliidine genera, particularly *Hyaliodes* Reuter, *Hyaliodocoris* Knight, *Knightonia* Carvalho and Drake, and *Paracarnus* Distant, are in need of revision.



Figs. 10.16–10.27 Neotropical Miridae. 16, *Diplozona collaris* Van Duzee (Deraeocorinae: Deraeocorini). 17, *Brasiliocarnus fraudans* (Carvalho) (Deraeocorinae: Hyaliodini). 18, *Bicuspidatiella conica* Maldonado (Deraeocorinae: Hyaliodini). 19, *Knightonia knighti* (Carvalho) (Deraeocorinae: Hyaliodini). 20, *Eustictus brunneus* Maldonado (Deraeocorinae: Surinamellini). 21, *Surinamella doesburgi* Carvalho and Rosas (Deraeocorinae: Surinamellini). 22, *Termatophylidea brunnea* Maldonado (Deraeocorinae: Termatophylini). 23, *Diphleps unica* Bergroth (Isometopinae: Diphlebini). 24, *Myiomma mexicanum* Henry (Isometopinae: Myiommini). 25, *Adxenetus petiolatus* (Stål) (Mirinae: Herdoniini). 26, *Calondas fasciatus* Distant (Mirinae: Mirini). 27, *Creontiades rubrinervis* (Stål) (Mirinae: Mirini) (Photos by G Ouellette)

Oliveira et al. (2002) studied *Hyaliodes beckeri* Carvalho (Hyaliodini) as a predator of the cassava pest *Vatiga illudens* (Heteroptera: Tingidae) in Brazil. Wheeler and Henry (2005) documented the association of *Bicuspidatiella conica* Maldonado with the formicine ant *Myrmelachista ramulorum* Wheeler (Formicidae) in Puerto Rico and speculated that this hyaliodine fed on the ant-attended mealybugs, soft scales, and whiteflies present on the host tree, laurel amarillo, *Nectandra turbacensis* (Kunth) Nees (Lauraceae).

Surinamellini (Figs. 10.20 and 10.21)

Five genera and 21 species of Surinamellini are known (Schuh 2013). The most important works for this group include descriptive papers by Carvalho (1988a) on *Craoiella* Carvalho, *Eustictus* Reuter, *Krainacoris* Carvalho and Wallerstein; Carvalho (1989b) on *Eustictus*, *Guapimirella* Carvalho; Carvalho and Rosas (1962) on *Surinamella* Carvalho and Rosas; and Carvalho and Costa (1990a, 1991) on *Eustictus*.

The habits of the surinamellines are poorly known. They are most often taken at lights or beaten from the branches of their hosts. Although there is little documentation, most or all members of this tribe probably are predatory, preying on small, coexisting arthropods. Most genera are in need of revision, and numerous undescribed taxa remain to be described.

Termatophylini (Fig. 10.22)

Only four genera and 10 species of this predatory tribe occur in the Neotropics (Schuh 2013). Maldonado (1970) reviewed the species of *Termatophylidea*, and Poppius and Carvalho (1955b) described new species of *Termatophyllella* Carvalho, *Termatophylidea*, and *Termatophyloides* Carvalho. Cassis (1995) presented a phylogenetic analysis and provided a checklist and key to the genera of the world.

Most species of *Termatophylidea* are thought to be thrips (Thysanoptera) specialists. Wheeler (2001) documented that *Termatophylidea pilosa* Reuter and Poppius (Myer 1935) preys on redbanded thrips (*Selenothrips rubrocinctus*) in Jamaica, and *T. maculata* Usinger and *T. opaca* Carvalho (Callan 1943) feed on redbanded thrips and grass thrips (*Caliothrips insularis*), respectively.

10.5.4 Subfamily Isometopinae (Figs. 10.23 and 10.24)

The Isometopinae are recognized by the overall round to elongate-oval body form; the trapeziform pronotum with carinate lateral margins; the strongly modified head, often with holoptic eyes, and distinct ocelli between the eyes; the sexually dimorphic antennae with segment II longer and more thickened in males than females; and the small size, ranging from 2.00 to 3.00 mm.

Herczek (1993) discussed the systematic relationships and position of the subfamily and recognized the two tribes Diphlebini and Myiommini in the New World. For the latter, he further recognized three subtribes, two of which occur in the Neotropical Region: Myiommina and Plaumanocorina.

Nine genera and 31 species are known from the Neotropics (Schuh 2013). Ghauri and Ghauri (1983) keyed the genera of the world, but overlooked several Western Hemisphere genera. Henry (1980) provided a key to the New World genera, reviewed the genera *Lidopus* and *Wetmorea*, and described the new genera *Brailovskiocoris*, *Lidopiella*, and *Myiopus*. Henry and Herring (1979) revised *Corticoris* McAtee and Malloch, Henry (1979c) described new species and provided a key to the New World species of *Myiomma* Uhler, and Henry and Carpintero (2012) treated the Isometopinae from Argentina, parts of Brazil, and Paraguay and described nine new species.

The genus *Isometocoris* Carvalho and Sailer (1954), first placed in Isometopinae, was transferred to Psallopinae by Henry and Maldonado (1982). Wolski and Henry (2015) document the placement of this genus in the Cylapinae.

With the apparent exception of one Oriental species, evidence indicates that all Isometopinae are scale predators (Wheeler and Henry 1978a; Henry 1984b; Wheeler 2001).

10.5.5 Subfamily Mirinae

The Mirinae is recognized by the distinct pronotal collar, three-segmented tarsi, divergent fleshy parempodia, inflatable endosomal membrane, and the large, ringed secondary gonoporal opening.

The following key is modified from Hernández and Henry (2010).

Key to the Tribes of Mirinae

1. First segment of hind tarsus as long as second and third combined; antennae usually very long and slender; head porrect; frons usually with a distinct median sulcus; coloration brown or green Stenodemini
- First segment of hind tarsus shorter than second and third combined; antennae relatively short; frons without a median sulcus; coloration variable 2
2. Species mostly dull black, marked with red, orange, or yellow on head, pronotum, and/or scutellum, dorsal surface velvety; metathoracic evaporative area small; pronotal collar wide mesally Restheniini
- Species variably colored, usually green, brown, or black with variable markings, dorsal surface dull or shiny, but never velvety; metathoracic evaporative area prominent; pronotal collar more narrow 3
3. Strongly myrmecomorphic; abdomen constricted at base in most taxa, hemelytra usually constricted through middle, corium usually with pruinose patches and often with a white spot near middle margin Herdoniini
- Not strongly myrmecomorphic; abdomen never constricted at base; hemelytra usually subparallel or convex, usually without pruinose patches and white spots Mirini

Herdoniini (Fig. 10.25)

Twenty-seven genera and 78 species of this group of strikingly ant-mimetic taxa are known from the Neotropics (Schuh 2013). Carvalho (1973a) provided a list of the genera, and Carvalho (1973b) keyed the genera of the world. In a series of well-illustrated papers, Carvalho and Ferreira (Carvalho and Ferreira 1973a, b, c, d, e, f, g, h, i, j, k, l) treated most of the herdoniine genera, usually accompanied by a key to species: *Adxenetus* Carvalho and Ferreira, *Allommatisca* Carvalho and Ferreira, *Allommatus* Reuter, *Barberiella* Poppius, *Camponotidea* Reuter, *Cearana* Carvalho and Ferreira, *Dacerla* Bergroth, *Guarania* Carvalho and China, *Haarupia* Poppius, *Herdonisca* Carvalho and Ferreira, *Herdonius* Stål, *Laurinia* Reuter, *Lepidoxenetus* Poppius, *Paradacerla* Carvalho and Usinger, *Paraxenetus* Reuter, *Proxenetus* Carvalho and Ferreira, *Sphinctothorax* Stål, *Xenetomorpha* Poppius, *Xenetopsis* Poppius, *Xenetus* Distant, *Zacynthus* Distant, and *Zosippus* Distant.

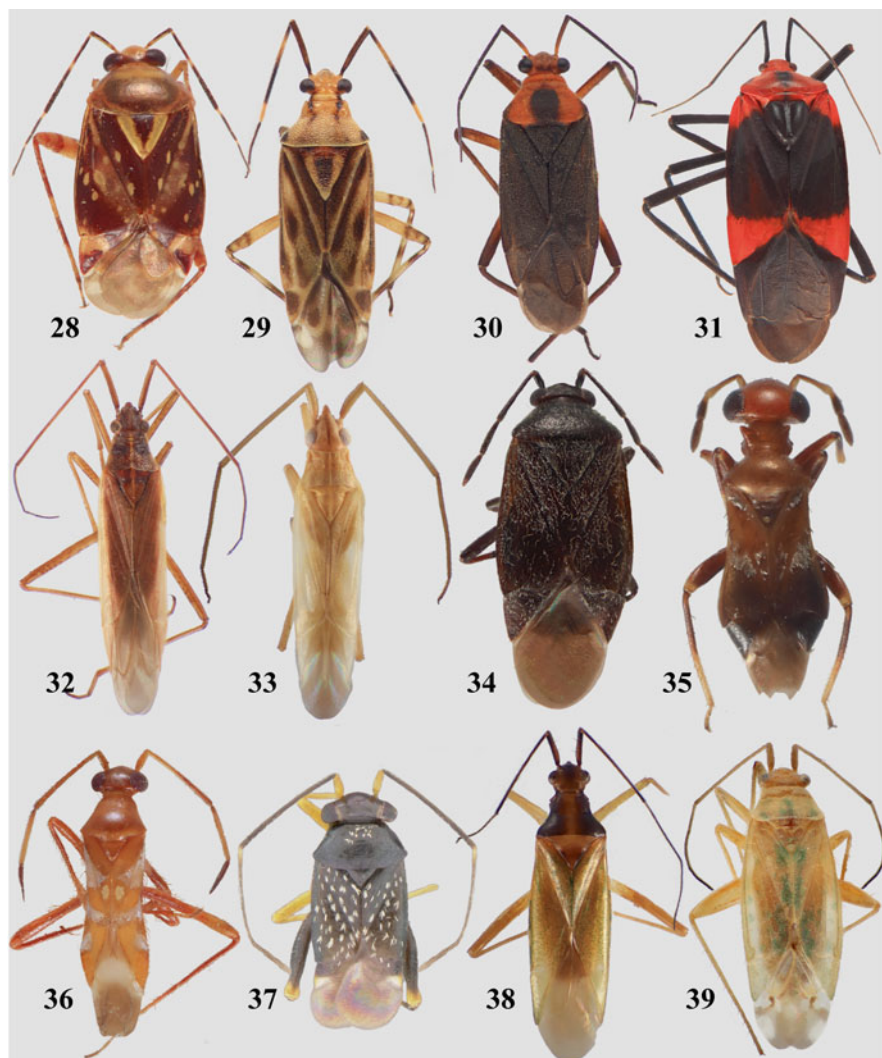
Herdoniines are remarkable in their resemblance to ants, with most having the abdomen constricted at the base and hemelytra narrowed through the middle. Wheeler and Henry (1980) studied *Barberiella formicoides* (Poppius) in the eastern United States and documented its habits and intimate association with three different species of ants. Little other information is available on the habits and life history of this fascinating group.

Mirini (Figs. 10.26, 10.27, 10.28, and 10.29)

Eighty-seven genera and more than 600 species of Mirini occur in the Neotropics. Carvalho's (1955a) key remains the only attempt to identify all genera of Mirini.

Phytocoris Fallén, the largest genus within the Miridae (Stonedahl 1988b), is represented in the Neotropics by 94 species (Schuh 2013), the second largest number after *Prepops* Reuter (Restheniini). Stonedahl (1988b) revised and keyed the western North American species, many of which range into Mexico, but few studies treat the Neotropical fauna much beyond describing new species. Other useful works on this genus include Maldonado (1969), Carvalho and Ferreira (1969), Carvalho and Gomes (1970), Carvalho (1990b), Carvalho and Costa (1990b, d, 1995), Carpintero and Chérot (2008, 2011), and Hernández and Henry (2010).

Important revisions of the largest genera, with keys to species, include Carvalho (1986a) for 12 species of *Calocorisca* Distant; Carvalho (1986b) for 12 species of *Chrysodasia* Carvalho; Carvalho and Fontes (1983) for 38 species of *Dagbertus* Distant; Carvalho and Gomes (1980), for 26 species of *Derophthalma* Berg; Carvalho (1986c) for 13 species of *Euchilocoris* Reuter; Carvalho (1976a) for 34 species of *Horcias* Distant; Carvalho and Jurberg (1976) for 12 species of *Horciasinus* Carvalho and Jurberg; Carvalho and Fontes (1972) and Carvalho (1985c) for 29 species of *Neostenotus* Reuter; Carvalho and Ferreira (1971) and Fontes (1981) for 19 of *Notholopus* Bergroth; Carvalho (1975c) for 18 species of *Poeas* Distant; Carvalho and Gomes (1969) and Ferreira (1980) for 22 species of *Polymerus* Hahn; Carvalho and Costa (1989) for 20 species (South American spp.



Figs. 10.28–10.39 Neotropical Miridae. 28, *Horcias nobilellus* (Berg) (Mirinae: Mirini). 29, *Piasus cribricollis* (Stål) (Mirinae: Mirini). 30, *Prepops latipennis* (Stål) (Mirinae: Restheniini). 31, *Resthenia scutata* Spinola (Mirinae: Restheniini). 32, *Stenodema dohrni* (Stål) (Mirinae: Stenodemini). 33, *Trigonotylus tenuis* (Reuter) (Mirinae: Stenodemini). 34, *Izyacapsus kerzhneri* Henry (Orthotylinae: Ceratocapsini). 35, *Zanchisme mexicanus* Carvalho and Schaffner (Orthotylinae: Ceratocapsini). 36, *Sericophanes ornatus* (Berg) (Orthotylinae: Ceratocapsini). 37, *Microtechnites bractatus* (Say) (Orthotylinae: Halticini). 38, *Carvalhomiris bifurcatus* Forero (Orthotylinae: Orthotylini). 39, *Chileria colla* Carvalho and Carpintero (Orthotylinae: Orthotylini) (Photos by G Ouellette)

only) of *Proba* Distant; Carvalho and Costa (1993) for 60 species of *Taedia* Distant; and Carvalho (1986d) for 20 species of *Tropidosteptes* Uhler.

Many Mirini, such as species of the genus *Phytocoris*, are predatory (Stonedahl 1988b). Others, however, may become serious pests of agricultural crops. Many species of *Lygus*, including the ubiquitous tarnished plant bug, *L. lineolaris* (Palisot de Beauvois), cause millions of dollars worth of damage each year in the United States (Schwartz and Foottit 1992, 1998). Other mirines, such as the black grass bugs, *Irbisia* spp., are major pests of forage grasses in western North America (Schwartz 1984). Species of genus *Tropidosteptes* cause severe injury to *Fraxinus* and other genera of the plant family Oleaceae (Henry et al. 2012). In Mexico and Colombia, *T. chapingoensis* Carvalho and Rosas is a serious pest of ornamental street plantings of *Fraxinus chinensis* Roxb., which sometimes causes heavy chlorosis and defoliation (Fonseca-González et al. 2007).

Restheniini (Figs. 10.30 and 10.31)

Restheniines are strictly a New World group. They are aposematically colored bugs, usually red or orange and black. This tribe contains the world's largest known mirids, with some species of *Resthenia* more than 15 mm long (Henry 2009). Twenty-seven genera and 320 species of Restheniini occur in the Neotropics (Schuh 2013). Carvalho and Fontes (1971a) provided a key to genera.

Prepops Reuter represents the largest genus in the Neotropics with 175 described species. Much of the work on this group has been descriptive. No one has attempted to key all of the species, although Carvalho and Fontes (1969a) gave a partial key to 18 species. The most important papers treating this genus include Carvalho (1974a, 1987b), Carvalho and Fontes (1968b, 1969a, b, 1970a, b, 1971b, c, d, e, f), Fontes (1989), Coelho (2012), and Coelho et al. (2012).

Some of the more comprehensive works include Carvalho and Fontes' (1971e) study of *Chiloxionotus* Reuter (28 species) and Carvalho and Fontes' (1970c) and Carvalho and Schaffner's (1975) work on *Platytylus* Fieber (30 species). Some of the most descriptive publications include Carvalho and Carpintero (1987) (*Nanniresthenia* Carvalho, *Platytylus* Fieber, *Prepops*), Carvalho and Carpintero (1989) (*Carpinteroa* Carvalho, *Chiloxionotus*, *Mimoncopeltus* Kirkaldy, *Nanniresthenia*, *Platytylus*, *Prepops*), Carvalho and Ferreira (1968) (*Lampsophorus* Reuter, *Mabelia* Kirkaldy), and Carvalho (1974b, c) [*Lampsophorus*, *Mabelia*, *Prepops*, *Prepopsella* Carvalho, *Resthenia* Spinola, *Restheniella* Carvalho]; Carvalho (1975a) [*Chiloxionotus*, *Opistheurista* Carvalho, *Prepops*]; Carvalho (1987b, c, d) [*Carpinteroa*, *Eurylomata* Reuter, *Fontesius* Carvalho, *Guaranimiris* Carvalho, *Oncerometopus* Reuter, *Platytylus*, *Prepops*, *Seabracoris* Carvalho]; Carvalho (1988b, c) [*Chiloxionotus*, *Prepops*, *Resthenia*]; Carvalho (1989c) [*Carpinteroa*, *Mimoncopeltus*, *Platytylus*, *Prepops*, *Resthenia*].

Although the hosts and feeding habits of most restheniines are unknown, several species may cause economic damage to crops. *Opistheurista clandestina* (Van Duzee) causes extensive foliar spotting and blotching to beans, cowpeas, and sweet

potatoes in the southern United States and Central America (Wheeler 2001), resulting in injury resembling that caused by mites and thrips (Jones 1921). *Prepops latipennis* (Stål), another Neotropical species, produces similar injury to beans and potatoes (King and Saunders 1984).

Stenodemini (Figs. 10.32 and 10.33)

Schwartz (2008) revised the Stenodemini, provided a key to the genera of the world, and presented a cladistic analysis supporting the monophyly of this tribe of grass-feeding bugs. Thirteen genera and about 46 species are known from the Neotropics (Schuh 2013).

Carvalho and Fontes (1981) reviewed the Neotropical species of *Collaria* Provancher, Carvalho (1975b) keyed the species of *Stenodema* Laporte, and Carvalho and Fontes (1969c) treated *Neotropicomiris* Carvalho and Fontes. Other mostly descriptive contributions include Carpintero and Estevez (2001) (*Cynodonmiris* Carpintero and Estevez, *Spartinomiris* Carpintero and Estevez), Carvalho and Fontes (1969c) (*Collaria*, *Dolichomiris* Reuter, *Kuscheliana* Carvalho, *Neotropicomiris*, *Ophthalmomiris* Berg, *Porpomiris* Berg, *Stenodema*, *Trigonotylus* Fieber), and Carvalho (1975b) [*Dolichomiris*, *Megaloceroea* Fieber, *Stenodema*, *Trigonotyliscus* Carvalho (a junior synonym of *Trigonotylus*), *Trigonotylus*].

Losses to prairie grasses, important graze for cattle and milk production in Colombia, have been attributed to stenodemines. *Collaria scenica* (Stål), the predominant pest species in that region, has caused losses up to 25 % of dry matter, the decrease of 10 % grass digestibility, and the reduction of milk production by up to 5 l/animal/day, resulting in an economic loss of 20 % to producers (Martinez and Barreto 1998; Duarte et al. 1998; Barreto 2011). Its biology has been studied on wheat, including adult morphometry (Carlessi et al. 1999).

Collaria oleosa (Distant) is a potential pest of wheat in Brazil (Da Silva et al. 1994) and is known to reduce sorghum yields in Cuba (Ryder et al. 1968). Ferreira et al. (2001) evaluated the economic importance of *Collaria oleosa* (Distant), *Dolichomiris linearis* Reuter, *Neotropicomiris costalis* Carvalho and Fontes, *N. longirostris* Carvalho and Fontes, *Opisthocoris carmelitanus* Carvalho and Costa, and *Trigonotylus tenuis* Reuter on grasses in Minas Gerais, Brazil.

10.5.6 Subfamily Orthotylinae

The Orthotylinae are distinguished by the usually complex male genitalia, especially the endosoma, often with multiple branching spicules, and parameres; the relatively large, open genital aperture; and the fleshy, convergent parempodia between the claws that are similar to those found in pilophorine Phylinae (distinguished, however, by the strap-like vesica similar to all other phylines).

The following key is modified from Hernández and Henry (2010).

Key to the Tribes of Orthotylinae

1. Small, oval, compact species with saltatorial hind femora 2
 - Usually larger, more elongate, medium-sized species, hind femora not saltatorial 3
2. Black species; head strongly dorsoventrally elongate; dorsum usually with patches of white, scalelike setae; parameres small, simple Halticini
 - Pale yellow to brown species; head not strongly dorsoventrally elongate; dorsum without scalelike setae; left paramere, long, sickle shaped Coridromini
3. Head truncate posteriorly, with a distinct basal carina; species often ant mimetic but, if not, antennal segments III and IV as thick or thicker than antennal segment II; male parameres sickle shaped to complex and branching; vesica without or, at most, with one simple spiculum; phallosome readily visible within genital capsule when viewed caudally Ceratocapsini
 - Head not truncate posteriorly, without a distinct basal carina; species not ant mimetic; antennal segments III and IV more slender than segment II; male parameres relatively simple; vesica with multiple, often branching or complex spiculi; phallosome not visible within genital capsule Orthotylini

Ceratocapsini (Figs. 10.34, 10.35, and 10.36)

The Ceratocapsini are an exclusively New World group, comprising more than 10 genera (Henry 1979b, 1994, 2006, 2013) and about 200 species, about half of which occur in the Neotropics. Many ceratocapsines are remarkably myrmecomorphic and are frequently collected in association with ants (Wheeler 1991; Henry 1994). Carvalho (1958b), in his world catalog, synonymized the tribe with the nominate tribe Orthotylini. Since then, however, Carvalho et al. (1983) and Henry (e.g., 1994, 2006, 2013) have given evidence for its monophyly, and, more recently, Schuh (2013) recognized the tribe in his online catalog.

Carvalho et al. (1983) reviewed the species of *Ceratocapsus* Reuter for the Neotropical Region, described 45 new species, and provided a key to distinguish them. Carvalho and Costa (1988) reviewed the genus *Sericophanes* Reuter and provided a key to the Neotropical species, and Hernández and Henry (1999) treated the Cuban species of *Ceratocapsus*. Since then, most papers have been primarily descriptive. More recently, Henry (2006) added the new genus *Izyacapsus* to accommodate two new species from Mexico, Henry (2013) added the new genus *Joseocoris* to include two new species from Argentina and Brazil, and Henry (2015) revised the *Renodaues* group of species, comprising eight genera and 48 species, including 26 new species, many of which are Neotropical.

Ceratocapsines are largely predatory. Wheeler and Henry (1978b) documented *Ceratocapsus modestus* (Uhler) preying on grape phylloxera, *Daktulosphaira viti-*

foliae (Fitch), in North America. Carvalho et al. (1983) provided an overview of the feeding habits of the group, including a report of *C. dispersus* Carvalho and Fontes feeding on the tobacco budworm *Heliothis virescens* (F.) in Peru. Gravena and Pazetto (1987) observed *C. dispersus* Carvalho and Fontes and *C. mariliensis* Carvalho and Fontes preying on the eggs and larvae of the cotton leafworm *Alabama argillacea* (Hübner) and tobacco budworm in Brazil. Wheeler (2000b, 2001) provided a summary of ceratocapsine feeding habits.

Coridromini

This newly recognized tribe was established for the Old World genus *Coridromius* Signoret (Tatarnic and Cassis 2012), based on hypodermic genitalia formed by the left paramere and aedeagus, the reduced female genitalia, the short round body, and strongly saltatorial hind femora (Tatarnic et al. 2006).

Only the adventive *Coridromius chenopoderis* Tatarnic and Cassis (2008) occurs in Neotropical Region. In addition to the Australian Region, it is now known from Hawaii, the southwestern United States (and Florida), and Mexico, where it feeds on plants in the family Chenopodiaceae (Tatarnic and Cassis 2008).

Halticini (Fig. 10.37)

The halticines are a group of mostly small, shiny, black mirids, with a dorsoventrally elongate head, often with silvery scalelike dorsal setae set in scattered patches, simple parameres, and a mostly membranous endosoma with two sclerotized spiculi (Tatarnic and Cassis 2012).

Only the genus *Microtechnites* Berg and five species are currently recognized from the Neotropics (Schuh 2013). Tatarnic and Cassis (2012) transferred most New World species of *Halticus* Hahn to *Microtechnites* and placed *Cafayatina* Carvalho and Carpintero as a junior synonym of it. Henry (1983) reviewed the New World *Halticus*, several of which are now in *Microtechnites*, and provided a key to distinguish species.

The ubiquitous garden flea hopper, *Microtechnites bractatus* (Say) (widely known in the literature under the combination *Halticus bractatus*), occurs throughout much of the Western Hemisphere from Canada to Argentina and Brazil (Henry and Wheeler 1988), where it may cause significant injury to agricultural crops. Beyer (1921) and Cagle and Jackson (1947) studied the life history. This species may cause severe foliar chlorosis or bleaching, often resulting in heavy crop losses. It has a wide host range, including small grains and many forage and garden crops; major losses to alfalfa, clover, and tomatoes have been recorded (Beyer 1921; Morrill 1925).

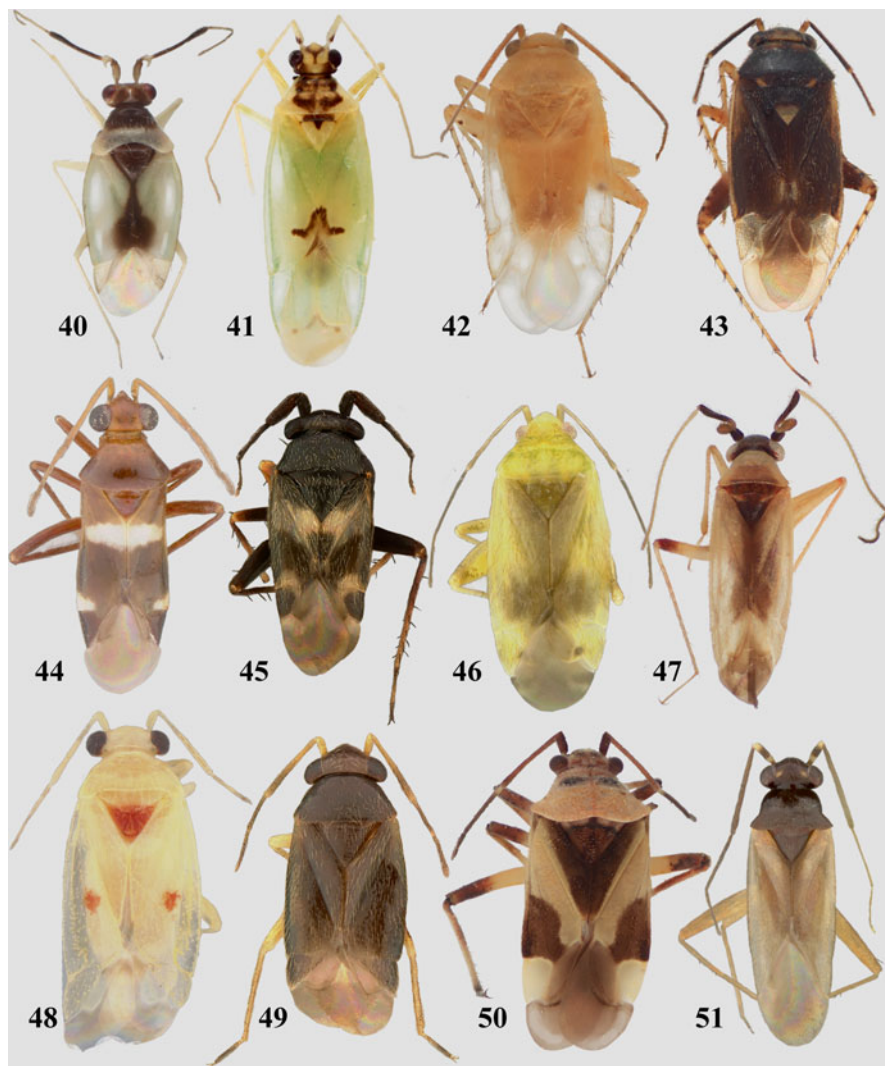
Orthotylini (Figs. 10.38, 10.39, 10.40, and 10.41)

Carvalho (1985b) provided a key to genera of Neotropical Orthotylinae.

Other important contributions to the study of this subfamily in the Neotropics are Forero (2008) (*Aoplonema* Knight, *Aoplonemella* Forero, *Daleapidea* Knight, *Hadronema* Uhler, *Hadronemidea* Reuter, *Lopidea* Uhler, *Origonema* Forero, *Scutomiris* Forero, *Tupimiris* Carvalho and Schaffner); Forero and Schwartz (2009) (*Aoplonema*); Forero (2009) (*Chileria* Carvalho, *Orthotylus* Fieber); Henry (1978, 2001) (*Hyalochloria* Reuter); Knight and Schaffner (1968) (*Lopidea*, *Mayamiris* Knight and Schaffner), 1975 (*Lopidea*), 1976 (*Ilnacora* Reuter); Maldonado and Ferreira (1971) (*Carvalhomiris* Maldonado and Ferreira); Maldonado (1980) (*Jobertus* Distant); Schaffner (1969) (*Lopidella* Knight); and Schaffner and Schwartz (2008) (*Ficinus* Distant, *Jornandes* Distant).

The habits of the Orthotylinae are widely varied. Numerous species may become pests of various crops and ornamentals. For example, Wheeler and Henry (1976) documented the life history and damage by the honey locust plant bug, *Blepharidopterus chlorionis* (Say), on ornamental honey locust, *Gleditsia triacanthos* L. (Fabaceae), in the United States. Species of the onion plant bug genus *Labopidea* Uhler may seriously injure commercial onions and garlic (Henry 1982), and *Falconia intermedia* (Distant) is known to discolor *Lantana camara* L. (Verbenaceae) in Mexico (Palmer and Pullen 1998), making it a potential biological control agent. Ferreira et al. (2001) and Pires et al. (2007, 2008) highlighted the feeding habits and life history of *Platyscytus decempunctatus* on its only known host plant, *Solanum cernuum* Vell. (Solanaceae).

A great number of other orthotylinae are predatory and have potential as biocontrol agents. The Holarctic *Blepharidopterous angulatus* (Fallén), known as the black-kneed capsid, is a well-documented predator of aphid and mites in orchards and ornamental plantings (Wheeler and Henry 1992). Hsiao (1945) described *Hyalochloria denticornis* from specimens taken on cotton and beans heavily infested with a species of *Empoasca* (Cicadellidae), suggesting that it was attracted to these leafhoppers as prey, and Beingolea (1959, 1960) observed this mirid feeding on *Anomis texana* Riley (Lepidoptera) and *Aphis gossypii* Glover (Aphididae) on cotton in Peru. Henry (2001) speculated that members of this genus are opportunistic predators, tracking their prey across multiple and often unrelated hosts. Numerous other members of this tribe, such as *Heterotoma planicornis* (Pallas), *Orthotylus ramus* Knight, and *O. marginalis* Reuter, have been observed preying on aphids, scales, and other small arthropods (Wheeler 2001), making it plausible that many of the poorly known Neotropical taxa should have similar predatory tendencies.



Figs. 10.40–10.51 Neotropical Miridae. 40, *Hyalochloria bispina* Henry (Orthotylinae: Orthotylini). 41, *Jobertus chryselectrus* Distant (Orthotylinae: Orthotylini). 42, *Hyalopsallus diaphanus* (Reuter) (Phylinae: Exaeretini). 43, *Moissonia cuneata* (Stål) (Phylinae: Exaeretini). 44, *Cyrtopeltocoris albofasciatus* Reuter (Phylinae: Hallodapini). 45, *Spanagonicus albofasciatus* (Reuter) (Phylinae: Nasocorini). 46, *Reuteroscopus hamatus* Kelton (Phylinae: Phylini: Keltoniina). 47, *Anomalocornis peyreti* Couturier and Costa (Phylinae: Phylini: Phylina). 48, *Platyscytus binotatus* Reuter (Phylinae: Phylini: Phylina). 49, *Sthenaridea vulgaris* (Distant) (Phylinae: Pilophorini). 50, *Semium subglaber* Knight (Phylinae: Semiini: Semiina). 51, *Tytthus neotropicalis* Carvalho (Phylinae: Semiini: Semiina) (Photos by G Ouellette)

10.5.7 Subfamily Phylinae

This subfamily is recognized by the hairlike parempodia in most taxa (except Pilophorini and some Hallodapini), the fleshy pulvilli often found along the ventral surface of each claw, the mittlike left paramere that usually cups the phallosheca, and the strap-like vesica of the male genitalia. Schuh (1974) noted that males are always macropterous, whereas females sometimes may be brachypterous.

Schuh (1995), in his catalog of the Miridae of the world, recognized five phylina tribes (Aurillocorini, Hallodapini, Leucophoropterini, Phylini, and Pilophorini) and, subsequently, listed six tribes in his online catalog (Schuh 2013), following Wyniger's (2010) resurrection of Knight's tribe Pronotocrepini. These studies were followed by Schuh and Menard (2013), who significantly revised the classification and proposed eight tribes, based on a combination of molecular and morphological evidence, and Menard et al. (2014), who further redefined the classification in a "total-evidence phylogenetic analysis," proposed the tribes Coatonocapsini, Cremnorrhini, Decomiini, Hallodapini, Leucophoropterini, Nasocorini, Phylini, Pilophorini, and Semiini.

Only six of these nine phylina tribes are known from the Neotropics. Unfortunately, several of the newly proposed taxa lack meaningful morphological support, so that they cannot be readily keyed or defined without supporting molecular data, and genera can be placed in their respective tribes and subtribes only by following Schuh and Menard's (2013) listing. As a consequence, we provide an overview of the tribes as defined by Schuh and Menard (2013) and Menard et al. (2014), but are unable to provide a key to distinguish them until these taxa are defined morphologically.

Cremnorrhini

Schuh and Menard (2013) and Menard et al. (2014) recognized this tribe based exclusively on molecular data, indicating that unifying morphological characters are yet to be defined. Two subtribes within this tribe were recognized (Schuh and Menard 2013), the Old World Coatonocapsina and the mostly north temperate Cremnorrhina, the latter of which includes all taxa occurring in the Western Hemisphere. Subsequently, Menard et al. (2014) elevated the Old World Coatonocapsini to tribal status.

Most members of this tribe occur in the western United States and Canada, with only six species of *Coquilletia* Uhler, one of *Teleorhinus* Uhler, and one species of the large Holarctic genus *Macrotylus* Fieber recorded from Mexico (Henry and Wheeler 1988; Wyniger 2010, 2012). Most are recognized by their elongate, often ant-like body, the frequently brachypterous females, the oblique head with the clypeus visible dorsally, the pronotum with an upturned collar-like area, the enlarged free pulvilli, and the distinctive male and female genitalia (Wyniger 2010; Schuh and Menard 2013). Wyniger (2010, 2012) revised and provided identification keys

to the New World taxa under the tribal name Pronotocrepini, now treated as a synonym of Cremonorrhina by Schuh and Menard (2013).

Exaeretini (Figs. 10.42 and 10.43)

Only three Neotropical genera and about seven species are included in this tribe (Schuh and Menard 2013), including *Gonoporomiris* Henry and Schuh, *Hyalopsallus* Carvalho and Schaffner, and the circumtropical genus *Moissonia* Reuter. Schuh and Menard (2013) defined this tribe based on the relatively long labium extending beyond the hind coxae, the elongate claws without or with only very small pulvilli, a large twisted gonopore, a twisted S-shaped endosoma composed of two straps connected by a membrane, and an elevated posterior lobe on the left paramere.

Gonoporomiris contains two species (Henry and Schuh 2002; Schuh and Schwartz 2004), *Hyalopsallus* one (Henry and Wheeler 1982; Carvalho 1990a), and *Moissonia* four Neotropical species (Schuh 1995). Little is known of their habits, but they are probably phytophagous. Henry and Wheeler (1982) recorded adults and nymphs of *H. diaphanus* (Reuter) from *Crotalaria incana* L. (Fabaceae) in Florida, and Henry and Schuh (2002) reported adults of *Gonoporomiris mirificus* (Distant) from the flowers of cabbage palmetto, *Sabal* sp. (Arecaceae) on Grand Bahama Island.

Hallodapini (Fig. 10.44)

The Hallodapini are mostly Old World and include taxa previously placed in the Auricillocorini, a tribe recently synonymized with it by Schuh and Menard (2013). Hallodapines are recognized by the uniquely shaped flattened pronotal collar, projecting frons and clypeus, and the twisted endosoma. *Cyrtopeltocoris* Reuter, the only hallodapine genus represented in the Western Hemisphere, occurs primarily in the western United States, with one species known from Cuba and two ranging into Baja California, Mexico. Most species of *Cyrtopeltocoris* are shiny, dark brown, with distinct white maculae or spots on the hemelytra. Knight (1968) provided a key to distinguish eight of the eleven known species; Hernández and Henry (2010) redescribed *C. cubanus* Poppius.

Little is known of their habits. Females are brachypterous and extremely ant-like. Males are most frequently taken at lights.

Nasocorini (Fig. 10.45)

Ten genera and about 63 species of this tribe occur in the Neotropics (Schuh and Menard 2013), including the widespread genera *Atractotomus* Fieber, *Rhinacloa* Reuter, *Chlamydatus* Curtis, and *Spanagonicus* Berg. Schuh and Menard (2013) defined the Nasocorini based on the flattened vertex of many species; the short,

wide hind femora, often with a row of dorsal spicules on the hind femur; the sericeous or scalelike setae; and the cushion-like pulvilli covering the undersurface of the claws.

Recent revisions and keys to species of the Holarctic genera *Atractotomus* (Stonedahl 1990) and *Chlamydatus* (Schuh and Schwartz 2005), and the New World genera *Rhinacloa* (Schuh and Schwartz 1985) and *Spanagonicus* Menard (*in press*) will allow recognition of most species.

Members of this tribe are primarily phytophagous, but many have predatory tendencies. Stonedahl (1990) discussed the feeding habitus of the species of *Atractotomus*, noting that members of the *magnicornis* group probably were partially predacious. Wheeler (2001) summarized the predatory habits of *A. magnicornis* (Fallén) and *A. mali* (Meyer-Dür), both of which are well-known predators of aphids, scales, mites, and other small arthropods.

Phylini

As noted by Schuh and Menard (2013), this tribe “continues to be an amalgam of genera lacking easily characterized unifying characters,” making it impossible to provide a diagnosis that will allow others to recognize the taxa included in this group. Schuh and Menard (2013) and Menard et al. (2014) recognized the subtribes Exocarpocorina (not in the Neotropics), Keltoniina, Oncotylina, and Phylina.

Keltoniina (Fig. 10.46)

Keltoniina contains one Afrotropical genus and three New World genera (Schuh and Menard 2013). This subtribe is recognized by the somewhat elongate head, small eyes; multiple types of vestiture, including long, erect, almost bristlelike setae, and clumps or patches of white sericeous setae; and the generally, stout, C-shaped endosoma.

The genus *Pseudatomoscelis* Reuter contains four species found from the southern United States to Panama and the West Indies, and *Keltonia* Knight includes 13 species, ranging from the southern United States and south to Colombia and Venezuela (Henry 1991). Henry (1991) revised and keyed the species *Keltonia* and *Pseudatomoscelis*, the latter of which includes the cotton fleahopper, *P. seriata* (Reuter), a well-known pest of cotton in the southern United States, and Henry (2002) added a new species of *Pseudatomoscelis* from the Dominican Republic.

The large genus *Reuteroscopus* Kirkaldy includes 54 species found from the United States to South America. The bright yellow body, marked with dark brown, is characteristic of most species, but the male endosoma with peculiar “feathered” appendages is unique in the Miridae. Kelton (1964) and Knight (1965) described and keyed most of the species, but the species limits, especially those described from females, are still unclear, reflecting the great need for revision.

Oncotylinina

Although molecular data unites the taxa included in this subtribe, no morphological characters define them (Schuh and Menard 2013; Menard et al. 2014).

Schuh and Menard (2013) placed nine Neotropical genera in this subtribe, with most ranging only into Mexico: *Hamatophylus* Weirauch (one species), *Knightophylina* Schaffner (1978) (one species), *Lineatopsallus* Henry (two species), *Maculimiris* Weirauch (two species), *Phymatopsallus* Knight (five species), *Schaffneropsallus* (one species), and *Viscacoris* Weirauch (four species).

Ranzovius Distant, with 10 described species occurring from the United States to southern Brazil (Henry 1999), is one of the more interesting phyline genera. All species are obligate spider commensals. Wheeler and McCaffrey (1984) studied the life history and association of the apparently kleptoparasitic *R. clavicornis* (Knight) (as *R. contubernalis* Henry) with the theridiid spider *Anelosimus studiosus* (Hentz); the bug appeared to feed mainly on small entrapped insects too small to attract the spider's attention and occasionally on dry cadavers. Henry (1984a, 1999) revised the genus and provided a key to species.

Phylina (Figs. 10.47 and 10.48)

The only morphological characters uniting this subtribe are the vaguely diagnostic long labium extending beyond the hind coxae and the absence of pronotal calli (Menard et al. 2014). Nevertheless, molecular evidence apparently always groups these taxa (Menard et al. 2014). Because of the lack of a morphological definition, all Neotropical genera placed in Phylina are treated as *incertae sedis* by Schuh and Menard (2013), indicating that much work remains to clarify the position of the taxa included in this group.

Thirteen genera and about 49 Neotropical species are placed in Phylina (Schuh and Menard 2013). Some of the included genera and the most important papers treating them are *Anomalocornis* Carvalho and Wygodzinsky (Couturier and Costa 2002), *Bicurvicoris* Carvalho and Schaffner (Carvalho and Schaffner 1973), *Crassicornus* Carvalho (Carvalho 1984), *Parafulvius* Carvalho (Costa and Couturier 2000), *Platyscytisca* Costa and Henry (Henry and Costa 2003), and *Platyscytus* Reuter (Maldonado and Carvalho 1981).

Pilophorini (Fig. 10.49)

This worldwide tribe, containing many myrmecomorphic taxa, is recognized by the fleshy convergent parempodia between the claws; the white sericeous or silky setae, often occurring in patches or distinct bands; and the carinate and upturned posterior margin of the vertex. Only three genera and about 11 species are known from the Neotropics.

Schuh and Schwartz (1988) treated the New World species of *Pilophorus* and the nine mostly Neotropical species of *Sthenaridea* Reuter, most of which are associated with grasses (Poaceae) and sedges (Cyperaceae). Schuh (1991) provided a phylogenetic, biogeographic, and host plant analysis of this tribe. Hernández and Henry (2010) described the new species *Pilophorus cubanus*, which is associated with *Pinus* (Pinaceae). Most species of *Pilophorus* are, at least partially predacious, feeding on aphids and other arthropods (Schuh and Schwartz 1988); Wheeler (2001) summarized the known hosts and feeding habits for some of the Nearctic and Palearctic species. More recently, Henry (2012c) described the peculiar new genus and species *Randallophorus schuhi* from Paraguay, which apparently is most closely related to *Sthenaridea* based on male genitalia.

Semiini (Figs. 10.50 and 10.51)

The tribe Semiini, as defined by Schuh and Menard (2013), contains two subtribes, the Exocarpocorina, with four Neotropical genera and six species restricted to Argentina and Chile, and the Semiina, with three genera and 16 species ranging from the United States to South America. The tribe is defined by the eyes being parallel to the anterior margin of the pronotum, the reduced scent gland opening, the sclerotized ventral sack in females, and the simple, elongate endosoma, with or without a well-developed secondary gonopore or a broad, flattened, T-shaped endosoma with a well-developed secondary gonopore.

Exocarpocorina

Only four genera [*Ampimpacoris* Weirauch (one species) and Schuh, *Araucanophylus* Carvalho (two species), *Chiloephylus* Carvalho (one species), and *Gonzalezinus* Carvalho (two species)] and six poorly known species of the subtribe are known in the Neotropics, all of which occur only in Argentina or Chile. All remaining genera are found in the Australian Region, making this the only transantarctic distribution in the Phylinae (Schuh and Menard 2013). These taxa are grouped based on unique asymmetrical sclerites of the female vestibulum, the elaborate posterior wall, the L-shaped phallosome, and the expanded central portion of the left paramere (Schuh and Menard 2013; Menard et al. 2014).

Semiina

Some of the better-known genera of the subtribe Semiina include *Semium* Reuter, with five Neotropical species (Kelton 1974), and *Tytthus* Fieber, with 11 species (Henry 2012b). All species of *Semium* are associated with species of *Chamaesyce* (Euphorbiaceae; older records from *Euphorbia* sp. apply to this genus) (Kelton

1974). Species of the genus *Tytthus* prey exclusively on the eggs of grass-feeding plant hoppers (Delphacidae), making them important biological control agents on agricultural monocots, such as rice and sugarcane (Henry 2012b).

10.6 Concluding Remarks

The Miridae, with more than 11,000 species, represent one of the 20 most hyperdiverse insect groups (Cassis and Schuh 2012). As their common name implies, many plant bugs are phytophagous and often become pests of agricultural crops and ornamental plants. *Lygus* bugs (*Lygus* spp.), the cotton fleahopper (*Pseudatomoscelis seriatus*), the cocoa capsids (*Monalonion* spp.), the orchid bugs (*Tenthecoris* spp.), the garden fleahopper (*Microtechnites bractatus*), and the bean capsid (*Pycnoderes quadrimaculatus*) are just a few examples of mirid plant pests that bring notoriety to the family through their feeding (Wheeler 2000a), which may be expressed as chlorosis, cankers, shot holing, stunting, fruit drop, yellowing and wilting foliage, and defoliation (Wheeler 2001). Many others have the potential to cause enormous economic losses even when symptoms are not readily apparent (Wheeler 2001).

Though mirids have not evolved the obvious morphological adaptations for capturing prey, such as raptorial front legs found in phymatine reduviids or stout, piercing-sucking mouthparts found in asopine pentatomids and most Reduviidae, many mirids, nevertheless, are successful predators. Most mirids will feed on the eggs, early instar larvae, nymphs, and dead or dying invertebrates (Wheeler 2001). Some taxa possess powerful salivary enzymes that immobilize their prey (Cohen 1996). Most members of the subfamily Deraeocorinae and many species of Mirinae (e.g., *Phytocoris* spp.), Orthotylinae (e.g., *Blepharidopterus angulatus*), Phylinae (e.g., *Campylomma verbasci*), and Bryocorinae (e.g., *Campyloneura virgula*, *Engytatus modestus*) have strong predaceous tendencies.

Despite their great importance both as plant pests and as predators, many new taxa remain to be described, especially in the Neotropical Region, and the biology, host plants, and phylogenetic relationships of most are unknown, except for the most agriculturally important taxa. Henry and Wheeler's (1988) speculation that the world fauna will approach 20,000 species once the Neotropical species are more fully studied likely is an underestimation. We hope this brief overview will stimulate much-needed additional taxonomic and biological work on this highly diverse, speciose group of economically important bugs.

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